Blessed G/Michael Catholic Secondary School

Semester 1 Biology Final Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Match the items under Column A with the correct descriptions under Column B.

1. Match the functional protein category with its specific biological example: "Contractile protein"

A. Actin
B. Enzymes
C. Serum albumin
D. Myoglobin
E. Hormones
F. Elastin

2. Match the functional protein category with its specific biological example: "Respiratory pigment"

A. Actin
B. Enzymes
C. Serum albumin
D. Myoglobin
E. Hormones
F. Elastin

3. Match the functional protein category with its specific biological example: "Chemical messengers"

A. Actin
B. Enzymes
C. Serum albumin
D. Myoglobin
E. Hormones
F. Elastin

4. Match the functional protein category with its specific biological example: "Transport protein"

A. Actin
B. Enzymes
C. Serum albumin
D. Myoglobin
E. Hormones
F. Elastin

5. Match the functional protein category with its specific biological example: "Structural protein"

A. Actin
B. Enzymes
C. Serum albumin
D. Myoglobin
E. Hormones
F. Elastin

II. Choose the best answer among the given alternatives.

____ **6.** A short segment of double-stranded DNA has 80 thymine and 80 guanine residues. What is the total number of nucleotides in this DNA fragment?

A. 160 **B.** 40 **C.** 320 **D.** 640

____ **7.** Which molecule carries specific amino acids to the ribosomes during protein synthesis?

A. m-RNA **B.** t-RNA **C.** r-RNA **D.** DNA

____ **8.** Which of the following is true about a phosphodiester linkage?

A. A 5' phosphate group of one nucleotide unit is joined to the 3' hydroxyl group of the next nucleotide
B. A 3' phosphate group of one nucleotide unit is joined to the 5' hydroxyl group of the next nucleotide
C. A 5' phosphate group of one nucleotide unit is joined to the 5' hydroxyl group of the next nucleotide
D. None of the above

____ **9.** The two complementary strands of a DNA double helix are held together by:

A. Covalent bonds **B.** Hydrogen bonds
C. Ionic bonds **D.** Phosphodiester bonds

____ **10.** Identify the purine base of nucleic acids among the following:

A. Cytosine **B.** Thymine **C.** Uracil **D.** Adenine

____ **11.** RNA serves as the primary genetic material in:
A. Some viruses and prokaryotes **B.** Some prokaryotes only
C. Some viruses only **D.** Some eukaryotes

____ **12.** Lipids are essential for the absorption and transport of fat-soluble vitamins such as:
A. Vitamin D **B.** Vitamin A
C. Vitamin K **D.** All of the above

____ **13.** Lipids play a major role in the formation of the insulating myelin sheath which surrounds:
A. Muscle fibers **B.** Nerve fibers
C. Blood vessels **D.** None of the above

____ **14.** Which of the following is NOT classified as a macronutrient?
A. RNA **B.** DNA
C. Salt (minerals) **D.** Protein

____ **15.** The polymerization reaction in which polysaccharides are synthesized from monosaccharides:
A. Requires the formation of glycosidic bonds between monosaccharides
B. Are hydrolytic reactions
C. Results in the condensation loss of water
D. Both A and C

____ **16.** Fats (animal) and oils (plant) are respectively rich in:
A. Saturated and unsaturated fatty acids
B. Unsaturated and unsaturated fatty acids
C. Saturated and saturated fatty acids
D. None of the above

____ **17.** Which of the following are major storage forms of carbohydrates in living organisms?
A. Starch **B.** Glycogen **C.** Cellulose **D.** Both A and B

____ **18.** Which monosaccharide sugars are present in sucrose (table sugar)?
A. Glucose and glucose **B.** Glucose and fructose
C. Xylose and glucose **D.** Glucose and galactose

____ **19.** Which of the following is a non-reducing sugar?
A. Glucose **B.** Sucrose **C.** Lactose **D.** Maltose

____ **20.** A carbohydrate found exclusively in mammalian milk is:
A. Fructose **B.** Lactose **C.** Maltose **D.** Sucrose

____ **21.** Triglyceride lipids are formed by the condensation reaction between:
A. Fatty acids and amines **B.** Fatty acids and glycerol (alcohols)
C. Fatty acids and amino acids **D.** Carbon and hydrogen

____ **22.** Which of the following represents a major biological function of lipids?
A. Long-term energy storage **B.** Steroid hormone synthesis
C. Structural component of cell membranes **D.** All of the above

____ **23.** Which of the following lipids is the most abundant storage lipid found in animal adipose tissues?
A. Triglycerides **B.** Phospholipids **C.** Waxes **D.** Sphingolipids

____ **24.** Which of the following lipoproteins is commonly referred to as "good cholesterol"?
A. LDL (Low-Density Lipoprotein) **B.** HDL (High-Density Lipoprotein)
C. Triglyceride **D.** Phospholipids

____ **25.** Omega-3 and omega-6 fatty acids are notable examples of:
A. Polyunsaturated fatty acids **B.** Saturated fatty acids
C. Monounsaturated fatty acids **D.** All of the above

____ **26.** Excessive dietary intake of saturated and trans lipids significantly contributes to the risk of:

A. Obesity **B.** Cardiovascular disease

C. Type 2 diabetes **D.** All of the above

____ **27.** All of the following are true characteristics of DNA EXCEPT:

A. They carry the code for cellular proteins
B. They degenerate immediately after the formation of proteins
C. They are chemically stable macromolecules
D. They exist as antiparallel double-stranded helices

____ **28.** Which of the following chemical components are present in every nucleotide monomer?

A. Pentose sugar **B.** Phosphate group

C. Nitrogen-containing base **D.** All of the above

____ **29.** The primary structure of a protein is held together and maintained by:

A. Peptide bonds **B.** Hydrogen bonds

C. Ionic bonds **D.** Disulfide bonds

____ **30.** A peptide molecule that contains two peptide bonds linking three amino acid residues is called a:

A. Dipeptide **B.** Tripeptide **C.** Tetrapeptide **D.** Polypeptide

____ **31.** What property causes phospholipid molecules to spontaneously arrange into a bilayer in an aqueous medium?

A. The length of their fatty acid chains
B. The presence of hydrophilic heads and hydrophobic tails (amphipathic nature)
C. Their high solubility in water
D. The presence of double bonds in fatty acids

____ **32.** How many molecules of water are released when 200 glucose monomers are joined by glycosidic bonds to form an unbranched polysaccharide?

A. 99 **B.** 200 **C.** 199 **D.** 198

____ **33.** The most abundant organic polysaccharide in nature is:

A. Cellulose **B.** Starch **C.** Glycogen **D.** Chitin

____ **34.** Which complex structural polysaccharide forms the cell wall of fungi and arthropod exoskeletons?

A. Cellulose **B.** Starch **C.** Glycogen **D.** Chitin

____ **35.** During which phase of the eukaryotic cell cycle does DNA replication take place?

A. G1 phase **B.** G2 phase **C.** S phase **D.** M phase

____ **36.** What is the correct chronological sequence of the main phases of the eukaryotic cell cycle?

A. G1 -> G2 -> S -> Mitosis -> Cytokinesis
B. G1 -> S -> G2 -> Mitosis -> Cytokinesis
C. G1 -> G2 -> S -> Cytokinesis -> Mitosis
D. S -> G1 -> G2 -> Mitosis -> Cytokinesis

____ **37.** What key cellular event occurs during the anaphase stage of mitosis?

A. Chromosomes condense and the nuclear envelope breaks down
B. Chromosomes align along the metaphase plate equator
C. Sister chromatids separate at centromeres and move toward opposite spindle poles
D. The cytoplasm divides into two separate daughter cells

____ **38.** Which eukaryotic cell organelle is responsible for organizing and forming the mitotic spindle fibers?

A. Centromere **B.** Centrosome

C. Kinetochore **D.** Cleavage furrow

____ **39.** The longest sub-phase of eukaryotic interphase is typically the:

A. G1 phase **B.** G2 phase **C.** S phase **D.** M phase

____ **40.** What is the primary mechanical role of the spindle apparatus during mitosis?

A. It makes the chromosomes visible under light microscopy
B. It chemically dissolves the nuclear envelope
C. It synthesizes duplicated copies of genomic DNA

D. It physically attaches to and separates chromosomes during division

___ **41.** Which of the following vital events take place during cell cycle interphase?

A. The cell grows and increases in volume
B. Nuclear genomic DNA replicates
C. Organelles duplicate and prepare for mitosis
D. All of the above

___ **42.** The shortest and quickest phase of mitotic division is:

A. Metaphase **B.** Anaphase **C.** Prophase **D.** Telophase

___ **43.** Suppose a human somatic cell with 46 chromosomes undergoes normal mitotic cell division. How many chromosomes will each daughter cell contain?

A. 23 **B.** 46 **C.** 92 **D.** 48